

AI Powered Fake News Detection and Source Traceability Using Blockchain



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Submitted for the partial fulfillment of BS Software Engineering degree to the

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APRIL, 2026

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1. Introduction/Background

The introduction of **fake news detection using blockchain** is given below

1.1 Problem Domain Overview

- The project belongs to the fields of Artificial Intelligence (AI), Natural Language Processing (NLP), and Blockchain Technology.
- AI and machine learning techniques are widely used for text classification and pattern recognition, which are useful for identifying fake news.
- Blockchain technology is used for secure, transparent, and tamper-proof storage of information, making it suitable for verifying news sources.

1.2 Background of the Problem

Social media platforms allow information to spread rapidly across the world.

- Unfortunately, this also enables the rapid spread of misinformation and fake news.
- Fake news can manipulate public opinion, influence elections, create panic, and damage the credibility of real journalism.
- Current verification methods rely mostly on manual fact-checking, which is slow and inefficient.

1.3 Real-World Relevance

- Fake news has become a global challenge affecting society, politics, and media credibility.
- Automated systems are needed to detect fake news quickly and verify the authenticity of information sources.
- Combining AI for detection and Blockchain for source traceability can provide a reliable solution to combat misinformation

2. Literature Review / Existing Systems

Fake news detection research uses both machine learning and deep learning techniques to identify false information online. Common machine learning models used in this field include Support Vector Machine (SVM) and Random Forest, while deep learning models such as CNN and LSTM are also widely applied. More advanced models like BERT are used to better understand language and context in news content.

These models are usually trained using datasets such as FakeNewsNet, LIAR, and Kaggle fake news datasets. Many studies have reported accuracy levels ranging from 90% to 97%, showing that these techniques are effective in detecting fake news. Researchers also use natural language processing (NLP) techniques such as text preprocessing, tokenization, and feature extraction to improve model performance. These techniques help the models better analyze the structure and meaning of news content.

However, despite high accuracy, these systems still face several challenges. Deep learning models often require high computational resources and longer training time. They also struggle to detect sarcasm, complex context, and misleading language in text. Additionally, many existing systems have limited capability to detect fake news in multiple languages or in multimedia formats such as images and videos.

The related works on related topic are given below:

Table No.1: Related Work

Systems	Accuracy	Dataset	Algorithm	Strenghts	Weakness
Fake News Detection Using Machine Learning Algorithms	92%	FakeNewsNet Dataset	Logistic Regression, Random Forest, SVMF	High accuracy	Poor performance on short news texts
A Survey of Fake News Detection	95%	COVID-19 Fake News Dataset	BERT Transformer	Large Datasets	Requires high computational resources
Fake News Using Machine Learning Techniques	90%	Kaggle Fake News Dataset	Random Forest, NLP	Social Media Detection	Struggles with sarcasm detection
The Spread of True and False News Online	94%	Twitter Fake News Dataset	Deep Neural Networks	Hybrid Models	Model biased toward large datasets
Detecting Opinion Spams and Fake News Using Text Classification	96%	FakeNewsNet Dataset	BERT	Context Understanding	Low interpretability
Fake news detection using knowledge graph	95%	FakeNewsNet Dataset	Graph Neural Networks	Knowledge Graphs	Complex model implementation

Deep Learning Based Fake News Detection in Online Social Media	94%	LIAR Dataset	CNN + LSTM	Transformer Models	Decreased performance
Fake News Detection Using Hybrid Machine Learning Model	95%	Fake News Net Dataset	Random Forest + SVMISOT	Automated Detection	Limited multilingual capability
AI-Based Fake News Detection Framework Using Machine Learning	93.5%	Fake News Net Dataset	Logistic Regression, Random Forest, SVM	Pattern Recoganization	Cannot detect multimedia fake news
Advanced Fake News Detection Using Transformer Models	97%	Kaggle Fake News Dataset	BERT + RoBERTa	Data Mining	High computational cost

2.1 Identified Research Gap

- Most existing systems focus only on **fake news detection**.
- Very few systems provide traceability of the original news source.
- Integration of AI detection with Blockchain-based source verification remains limited.

3.Problem Statement

The problem statement is given below:

- The rapid spread of fake news on social media and digital platforms makes it difficult for users to distinguish between authentic and misleading information.
- Existing fake news detection systems mainly focus on content analysis but fail to verify the credibility and origin of the information source.

- Therefore, there is a need for a system that not only detects fake news automatically using AI but also ensures transparency and traceability of news sources using Blockchain technology.
- Many online fake news articles do not clearly mention reliable source, making it difficult for the users to determine whether the information is authentic or misleading.
- Verifying news manually requires significant time and efforts, which makes it difficult to control the rapid spread of fake news on digital

4. Objective

The objective of the AI-Powered Fake News Detection project is to develop a system that can automatically identify whether a news article is real or fake using Artificial Intelligence (AI) and Natural Language Processing (NLP). The system analyzes the text of news articles and uses machine learning algorithms to classify the information accurately.

Another objective is to train the model on reliable datasets so it can learn patterns, writing styles, and language used in fake and real news. This project aims to reduce the spread of misinformation and help users verify the authenticity of news before sharing it online.

4.1 Main Objective

To develop an intelligent system that detects fake news using Artificial Intelligence and verifies the authenticity of news sources through Blockchain-based traceability.

4.2 Specific Objectives

- To collect and preprocess datasets containing real and fake news articles.
- To design and train machine learning or deep learning models for automatic fake news classification.
- To integrate Blockchain technology for secure storage and traceability of news sources.
- To develop a user-friendly system interface for analyzing and verifying news content

5. Proposed Solution / Methodology

The proposed solution for the AI-Powered Fake News Detection project is based on using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to analyze and classify news articles. First, a dataset containing both real and fake news will be collected from reliable sources such as FakeNewsNet or LIAR datasets. This dataset will be used to train and test the model.

In the next step, data preprocessing will be performed. This includes cleaning the text by removing stop words, punctuation, special characters, and irrelevant information. The text will then be converted into a structured format using techniques such as tokenization and vectorization so that it can be processed by machine learning models.

After preprocessing, machine learning or deep learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), or neural network models like CNN and LSTM will be applied. These models will learn patterns, writing styles, and linguistic features that differentiate fake news from real news.

Finally, the trained model will be integrated into a system where users can input a news article or headline. The system will analyze the text and classify it as real or fake with a predicted accuracy score. This methodology provides a fast, automated, and reliable solution for detecting fake news and reducing the spread of misinformation on digital platforms.

1.Data collection

- Gather datasets containing real and fake news articles from trusted sources.

2.Data Preprocessing

- Remove noise, punctuation, and irrelevant words.
- Perform tokenization, stop-word removal, and text normalization

3.Feature Extraction

- Convert textual data into numerical features using techniques such as TF-IDF or word embeddings.

4.Model Training

- Train machine learning models such as:
 - RoBERTa
 - Convolutional Neural Networks (CNN)
 - Long Short-Term Memory (LSTM)

5.Fake News Detection

The trained AI model analyzes news content and classifies it as Real or Fake.

6.Blockchain Source Traceability

- News source information is stored in a Blockchain ledger.
- Each news record is assigned a unique hash for verification.
- Blockchain ensures that source information cannot be altered or manipulated.

7.User Interface

- Users can input news text or headlines.
The system will display:
 - Fake/Real classification
 - Source verification status

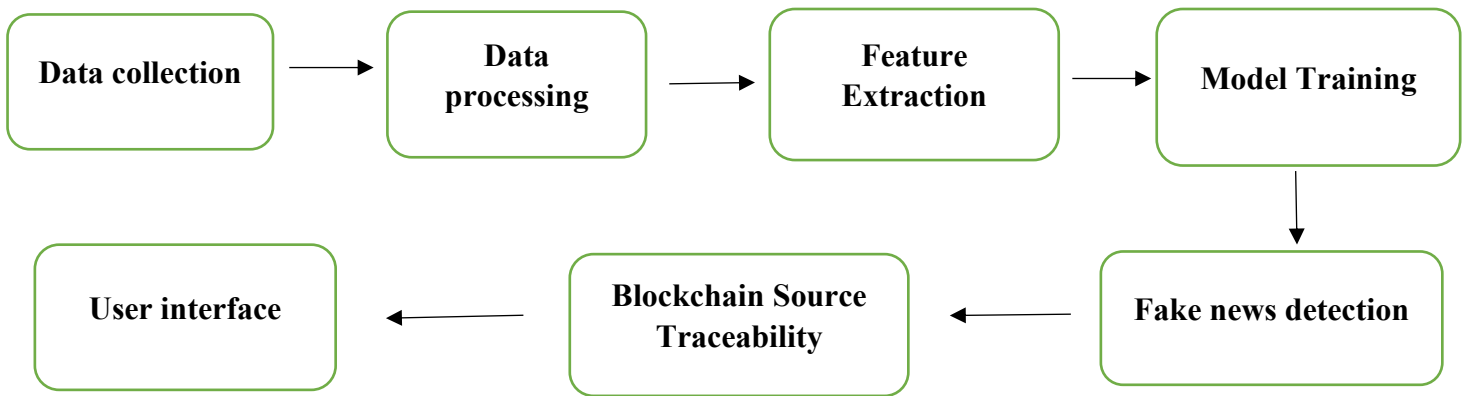


Figure 1:Workflow of AI based Fake News Detection

6.Tools and Technologies

6.1 Programming and Development Tools

Python

- Widely used for AI and machine learning development.

6.2Machine Learning Libraries

Scikit-learn

- Provides efficient machine learning algorithms for classification.

TensorFlow / PyTorch

- Used for implementing deep learning models.

6.3 Natural Language Processing Tools

NLTK

- Used for text preprocessing, tokenization, and language analysis.

6.4 Data Analysis

Pandas

- Used for data handling.

6.5 Blockchain Platform

Ethereum

- Used for creating a decentralized and secure ledger to store news source data.

6.6 Dataset Source

FakeNewsNet Dataset

LIAR Dataset

6.7 Frontend Development

Html, CSS, JavaScript

6.8 Interface/Backend Development

Flask / Web Application Framework

- Used to build a simple and interactive user interface.

7. Innovation / Novelty

The innovation of the AI-Powered Fake News Detection project lies in using Artificial Intelligence and Natural Language Processing to automatically analyze and classify news articles in real time. Unlike traditional fact-checking methods that rely on manual verification, this system provides a fast and automated way to detect fake news.

Another novel aspect is the use of advanced machine learning and deep learning models that can learn patterns in language, writing style, and context to improve detection accuracy. The system can continuously improve as it is trained with more data, making it more effective over time in identifying misleading or false information.

Moreover, the main innovations are given below:

- i. **Fake News Detection**
Existing systems rely only on basic AI classification.
Our approach improves accuracy using optimized AI with contextual analysis.
- ii. **Source Verification**
Traditional systems lack source verification.
Our system uses Blockchain for secure source traceability.
- iii. **Data Integrity**
Centralized systems are vulnerable to manipulation.
Our approach ensures tamper-proof storage using Blockchain.
- iv. **Transparency**
Existing systems provide limited validation insight.
Our system offers AI results with Blockchain-based verification.
- v. **Trust Mechanism**
AI-only systems lack full reliability.
Our approach uses a dual trust model (AI + Blockchain).
- vi. **System Architecture**
Traditional systems are centralized and less secure.
Our system is decentralized using Blockchain.

vii. **User Awareness**

Existing systems only classify news.

Our system provides classification with source traceability.

viii. **Security**

Current systems offer moderate protection.

Our approach ensures high security via cryptographic Blockchain mechanisms.

8. Social and Professional Benefits

The AI-Powered Fake News Detection project provides several social and professional benefits. Socially, it helps reduce the spread of misinformation on digital platforms and social media. By identifying fake news quickly, the system can help people make informed decisions and promote awareness about reliable information sources.

Professionally, the project demonstrates the practical use of Artificial Intelligence, Machine Learning, and Natural Language Processing in solving real-world problems. It also helps students and developers gain technical skills in data analysis, model training, and AI-based application development, which are valuable in the fields of data science and software development.

8.1 Social Benefits

The social benefits are given below

- Reduces the spread of misinformation and fake news on digital platforms.
- Promotes responsible sharing of information on social media.
- Helps users verify the authenticity of news sources.
- Improves public trust in digital news systems.

8.2 Professional Benefits

The professional benefits are given below:

- Provides hands-on experience in Artificial Intelligence, Machine Learning, and Blockchain technologies.
- Enhances research, data analysis, and software development skills.
- Contributes to the development of intelligent systems for information verification and cybersecurity.

9. Expected Outcomes

The outcomes of the results are given below:

- Development of a functional AI-powered Fake News Detection system integrated with Blockchain-based source traceability.

- The AI model is expected to achieve approximately 90% – 95% accuracy in classifying news articles as Real or Fake, depending on the dataset used.
- Expected Precision Score: around 88% – 93%, ensuring most detected fake news items are actually fake.
- Expected Recall Score: around 87% – 92%, enabling the system to correctly identify the majority of fake news instances.
- Expected F1-Score: approximately 89% – 94%, balancing precision and recall for reliable model performance.
- The Blockchain module is expected to provide 100% data integrity and traceability, ensuring that once a news source is recorded in the ledger, it cannot be altered or tampered with.
- The system will provide a user-friendly interface where users can:
 - Input news text or headline.
 - Receive instant classification results.

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